

M&S: Assignment 3D Robot Arm with cascaded control

Group 10:

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June 21, 2026

1 Introduction

This report details the integration of a 3-DoF spatial robot arm, combining realistic actuator dynamics, Series Elastic Actuation (SEA), and cascaded digital control. A primary focus of this project has been adopting a modular development methodology. This step-by-step approach allows for the isolated verification of individual subsystems before integration, mitigating the compounding errors we occasionally encountered in earlier weekly assignments.

Early Modelling Choices and Verification

An early and critical modelling choice was the decision to validate the analytical kinematics and dynamics against the 20-sim 3D Mechanics Editor. Initially, the arm performed well using simplified Motor-Spring-Equivalent (MSe) blocks. But we observed a persistent discrepancy: our analytical gravity compensation required a scaling factor of $K_g = 0.4$ to prevent steady-state error, which indicated a fundamental mismatch in the model. We discovered that the centers of mass (CoM) in our 3D model were misaligned relative to our analytical equations. By correcting the CoM placements in the 3D model to reflect the physical geometry the gravity compensation now operates with a theoretical gain of $K_g = 1.0$.

Focus Areas: Actuation, SEAs, and Discretisation

With the rigid body dynamics verified, our focus shifts to refining the actuation and control architecture:

- **Realistic Actuator Selection:** The motor-gearbox combination carried over from Assignment 2 was found to be either physically too bulky or dynamically insufficient for our 3D robot arm. We will focus on finding a better gearbox-motor combination.
- **SEA Retuning and Current Control:** The change in actuators will require retuning of our SEA parameters, particularly the viscous friction coefficients. In Assignment 5, we already upgraded the SEA control architecture from our more unconventional voltage-control-with-EMF-feedback approach to cascaded current and torque control loop; but we still need to discretise it.
- **Discrete-Time Implementation and cascaded control:** Given our limited prior experience with digital control, a larger portion of our effort will be dedicated to the correct implementation of discrete-time dynamics. We will focus on accurately modeling sampling delays, zero-order holds, and their impact on the stability margins of the cascaded PI controllers.

Simulation Experiments and Validation

To validate the integrated system, we designed a series of targeted simulation experiments:

1. **Isolated Subsystem Verification:** Testing the arm to verify gravity compensation (setting the first pitch actuator to 90° and see if this position can be maintained without further instruction), followed by isolated step-response tests of the retuned SEA and current control loops.
2. **Cartesian Impedance Interaction:** Executing a 3D motion task of picking up an object. This experiment validates the Cartesian impedance controller's ability to follow a trajectory and compensate for external forces.

2 Theoretical Background

2.1 3D-Robot Arm Dimensions

The 3D-robot-arm is modelled using three cylindrical links with a length of 250 mm and outer-radius of 25 mm. To see whether or not the dimensions were sensible, we made a CAD model and found the results sufficiently realistic. To match a realistic scenario material-wise, aluminum (Aluminum 6061-T6, $\rho = 2700$ ³) pipes were chosen to physically represent the model. The resulting mass for these links with outer radius 25 mm and inner radius 19 mm is:

$$V = \pi (R_{\text{out}}^2 - R_{\text{in}}^2) L = \pi [(0.025 \text{ m})^2 - (0.022 \text{ m})^2] \cdot 0.250 \text{ m} = 1.107 \cdot 10^{-4} \text{ m}^3$$

$$m = \rho \cdot V = 2700 \text{ kg/m}^3 \cdot 1.107 \cdot 10^{-4} \text{ m}^3 = \boxed{0.299 \text{ kg}}$$

The moments of inertia are:

$$I_{zz} = \frac{1}{2} m (R_{\text{out}}^2 + R_{\text{in}}^2) = \frac{1}{2} \cdot 0.299 \text{ kg} \cdot ((0.025 \text{ m})^2 + (0.022 \text{ m})^2) = 1.658 \cdot 10^{-4} \text{ kg} \cdot \text{m}^2$$

$$\begin{aligned} I_{xx} = I_{yy} &= \frac{1}{12} m [3 (R_{\text{out}}^2 + R_{\text{in}}^2) + L^2] = \frac{1}{12} \cdot 0.299 \text{ kg} \cdot [3 ((0.025 \text{ m})^2 + (0.022 \text{ m})^2) + (0.250 \text{ m})^2] \\ &= 1.640 \cdot 10^{-3} \text{ kg} \cdot \text{m}^2 \end{aligned}$$

Where I_{zz} is the longitudinal axis along the link and I_{xx} and I_{yy} are the transverse axes perpendicular to the link.

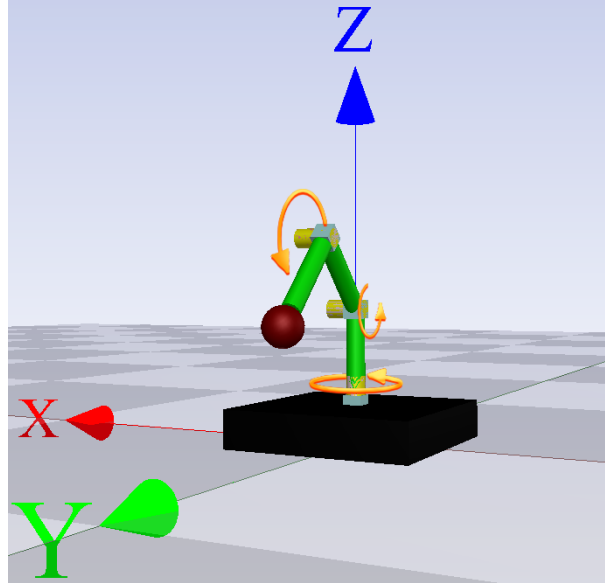


Figure 1: 3-DoF robot arm model

2.2 Actuator Selection and Sizing, verification using gravity compensation

For the series elastic actuation (SEA) of each joint in the 3-DoF robot arm, we selected the maxon EC-max 30 brushless DC motor (part-No.: 272766) coupled with the maxon GP-32-C planetary gearbox (part-No. 166951) with a ratio of 318:1. This combination was chosen over heavier alternatives for three primary reasons:

1. Low mass: Critical for minimising the inertia seen by proximal joints.
2. High reduction ratio (318:1): Enables high output torque from the small, fast-spinning motor.
3. Sufficient output speed: The motor's high no-load speed (8680 rpm) ensures sufficient joint speed after reduction.

The following parameters (tables 1 and 2) are used in our models (datasheets [1] and [2]):

Parameter	Value	Unit
Rated torque	0.0349	Nm
Peak (stall) torque	0.153	Nm
Rated speed	6630	rpm
No-load speed	8680	rpm
Rotor inertia	$1.10 \cdot 10^{-6}$	kg·m ²
Motor mass	0.195	kg
Torque constant	0.0129	Nm/A
Motor length	42.0	mm

Table 1: maxon EC-max 30 motor specifications

Parameter	Value	Unit
Reduction ratio, N	318	–
Efficiency, η	0.60	–
Gearbox inertia (input)	$7.0 \cdot 10^{-8}$	kg·m ²
Gearbox mass	0.226	kg
Gearbox length	49.8	mm

Table 2: Planetary gearhead GP32C (318:1) specifications, Part-No.: 166951

Important for the evaluation of the actuators are the values in table 3:

Quantity	Calculation	Result	Gearbox limits
Rated output torque	$\tau_{\text{out, rated}} = 0.0349 \cdot 318 \cdot 0.60$	6.66 Nm	6 Nm rated
Peak output torque	$\tau_{\text{out, peak}} = 0.153 \cdot 318 \cdot 0.60$	29.2 Nm	7.5 Nm max. intermittent
No-load output speed	$n_{\text{out, no-load}} = 8680/318$	27.3 rpm	✓
Total motor inertia	$J_{\text{total}} = 1.10e-6 + 7.0e-8$	$1.17e-6$ kg·m ²	–

Table 3: Performance summary with gearbox torque limits

To verify that the selected motor–gearbox combination provides sufficient torque, we used the gravity compensation equations as a baseline. To avoid excessive cantilevered inertias, the pitch actuators (Joints 2 and 3) were mounted in line with the local z-axes of the links. This arrangement achieves a "total zero-offset" for the lateral center of mass, ensuring the combined center of mass of each link-actuator assembly remains aligned with the joint axes.

To maintain computational efficiency while capturing the dominant gravitational effects, the actuator masses are simplified as point masses located at the center of rotation of each joint. Based on this simplification, the added gravity torques are evaluated as follows:

- **Joint 1:** The motor is aligned with the vertical Z-axis. Its mass creates no moment arm, resulting in negligible added gravity torque.
- **Joint 2:** The mass of Motor 2 is located directly at the joint axis, contributing no torque to Joint 2 itself. Furthermore, this joint does not deviate significantly from the vertical Z-axis during operation, keeping the gravitational moment arms relatively small. The primary added gravitational load on Joint 2 comes from Motor 3, which is mounted at the distal end of Link 2. This distal actuator is modeled as a point mass ($m_{\text{act}3} = 0.421$ kg) concentrated at the axis of joint 3.
- **Joint 3:** Motor 3 is located at the joint itself, adding no static gravity torque to Joint 3.

The gravity compensation torques, derived from the partial derivatives of the system's potential energy ($\tau_{gi} = \frac{\partial V}{\partial q_i}$), are formulated as:

$$\tau_{g1} = 0 \quad (1)$$

$$\tau_{g2} = -g \left[(m_2 r_2 + m_3 L_2 + m_{\text{act}3} L_2) \sin(q_2) + m_3 r_3 \sin(q_2 + q_3) \right] \quad (2)$$

$$\tau_{g3} = -g m_3 r_3 \sin(q_2 + q_3) \quad (3)$$

where g is the gravitational constant, r_2 and r_3 are the center-of-mass distances of the links, and $m_{\text{act}3}$ is the mass of the distal actuator. As shown in the equation, the mass of the third motor explicitly increases the static load on the second joint.

By evaluating worst-case gravity torques on paper (arm fully extended horizontally at 90°) and using the actual link parameters from our 3D robot arm model ($m_2 = m_3 = 0.331$ kg, $L = 0.25$ m, $r_2 = r_3 = 0.125$ m)

alongside the actuator mass m_{act3} , the resulting gravitational torque was found:

$$\tau_{g,2} \approx 2.25 \text{ Nm}, \quad \tau_{g,3} \approx 0 \text{ Nm (at the same configuration)}$$

The middle joint requires about 2.25 Nm to hold the arm still horizontally. The selected actuator provides a rated output torque of 6.6 Nm (limited by the gearbox to 6.0 Nm) and a peak torque of 29.2 Nm (limited by the gearbox to 7.5 Nm), comfortably exceeding the static gravity compensation demand.

Finally, these same parameters (including the correct centre-of-mass locations) were inserted into our 3D simulation model from assignment 7. The model returned almost identical gravity torques, confirming both the hand calculation and the actuator choice, and serving as an early verification of our gravity compensation formulation.

During testing it was later established that although the selected actuators provide enough torque for nominal trajectory tracking and minor disturbance rejection, the system lacks the torque margin required for “heavy lifting”. Boundary testing revealed that a 10 N external disturbance on the Z-axis at the end-effector requires 8.15 Nm of joint torque at the first pitch joint, which exceeds the actuator’s continuous limits. This is a physical hardware bottleneck in the current design.

Inertia of the actuator within the kinematics of the 3D robot arm

Approximating the combined actuator housing as a uniform solid cylinder with a diameter of 32 mm and an axial length of 70 mm (datasheet), the transverse moment of inertia (perpendicular to the joint axis), which dictates the resistance to being swung by proximal joints, is calculated as: The longitudinal moment of inertia (spin axis), which represents the inertia the motor must overcome to drive the pitch joint, is calculated as:

$$I_{xx} = \frac{1}{2} m_{act} r^2 = \frac{1}{2} (0.421) (0.016)^2 \approx 5.39 \cdot 10^{-5} \text{ kg m}^2$$

Conversely, the transverse moment of inertia (swing axis), which represents the resistance of the actuator housing to being swung through space by the proximal joints (e.g., the base yaw joint), is calculated as:

$$I_{zz} = I_{yy} = \frac{1}{12} m_{act} (3r^2 + L^2) = \frac{0.421}{12} [3(0.016)^2 + (0.070)^2] \approx 1.99 \cdot 10^{-4} \text{ kg m}^2$$

2.3 Kinematics

Given the Yaw-Pitch-Pitch (Z-Y-Y) kinematic structure (in our model it is configured as Z-X-X) of the 3-DoF arm, explicit analytical kinematics and gravity compensation were implemented in parallel with the 3D Mechanics Editor.

The position of the end-effector $\mathbf{p} = [x, y, z]^T$ in the base frame is derived from the joint angles $\mathbf{q} = [q_1, q_2, q_3]^T$, where q_1 is the base yaw, and q_2, q_3 are the shoulder and elbow pitch angles. Defining $q_{23} = q_2 + q_3$, the forward kinematics are:

$$x = \sin(q_1) (L_2 \sin(q_2) + L_3 \sin(q_{23})) \quad (4)$$

$$y = -\cos(q_1) (L_2 \sin(q_2) + L_3 \sin(q_{23})) \quad (5)$$

$$z = d_0 + L_1 + L_2 \cos(q_2) + L_3 \cos(q_{23}) \quad (6)$$

where L_2, L_3 are the link lengths, and $d_0 + L_1$ is the vertical offset of the shoulder joint.

The geometric Jacobian $\mathbf{J}(\mathbf{q}) \in \mathbb{R}^{3 \times 3}$, which maps joint velocities to end-effector Cartesian velocities ($\dot{\mathbf{p}} = \mathbf{J}\dot{\mathbf{q}}$), is obtained by differentiating the forward kinematics with respect to the joint angles:

$$\mathbf{J}(\mathbf{q}) = \begin{bmatrix} \cos(q_1)(L_2 \sin(q_2) + L_3 \sin(q_2 + q_3)) & \sin(q_1)(L_2 \cos(q_2) + L_3 \cos(q_2 + q_3)) & \sin(q_1)L_3 \cos(q_2 + q_3) \\ \sin(q_1)(L_2 \sin(q_2) + L_3 \sin(q_2 + q_3)) & -\cos(q_1)(L_2 \cos(q_2) + L_3 \cos(q_2 + q_3)) & -\cos(q_1)L_3 \cos(q_2 + q_3) \\ 0 & -L_2 \sin(q_2) - L_3 \sin(q_2 + q_3) & -L_3 \sin(q_2 + q_3) \end{bmatrix} \quad (7)$$

where $s_i = \sin(q_i)$ and $c_i = \cos(q_i)$. To map a desired Cartesian impedance force $\mathbf{F}_{imp} = [F_x, F_y, F_z]^T$ to the required joint torques $\boldsymbol{\tau}_{imp}$, the transpose of the Jacobian is utilized based on the principle of virtual work:

$$\boldsymbol{\tau}_{imp} = \mathbf{J}(\mathbf{q})^T \mathbf{F}_{imp} \quad (8)$$

In the current simulation, joint velocities $\dot{\mathbf{q}}$ are obtained directly from the 3D Mechanics Editor model. In a physical implementation, robots are typically equipped only with joint position encoders, and velocities have to be estimated through numerical differentiation. The direct velocity access in our simulation represents an idealised scenario that simplifies controller development.

To address this limitation, we implemented numerical differentiation for velocity estimation from joint positions, as shown in Figure 2. The results were indistinguishable from using the joint velocities extracted directly from the 3D model. In a physical implementation, position sensors would introduce measurement noise that gets amplified during differentiation, requiring low-pass filtering. Since our simulation does not include sensor noise, this issue does not arise, and there is no practical difference between the two approaches in our noise-free environment. But in order to scale our model for future implementations and in order to make it more realistic, we added the differentiation as depicted in figure 2.

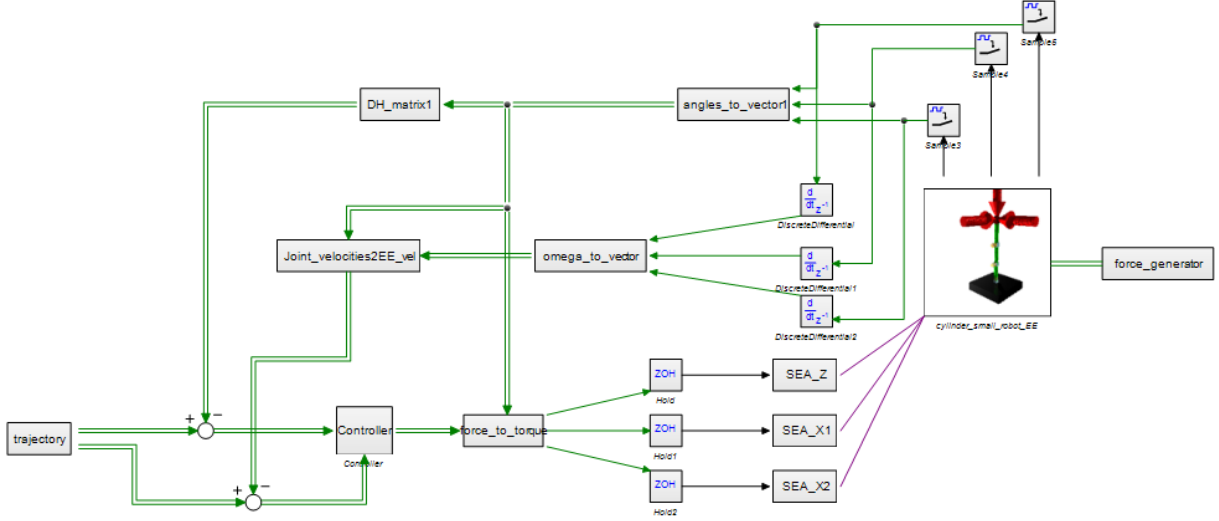


Figure 2: Full model using derivatives of the position encoders (file: diff_omega_model_circle_morestiff_test_discrete_SEAX2_SEAX1_SEAZ_PI.emx)

3 Model overview

The complete system model is hierarchically structured to separate the high-level Cartesian task space from the low-level joint actuation space. With this modular approach the different parts and submodels of the system can be verified independently.

We split our model up in these submodels:

1. **Cartesian Impedance Controller:** Computes the desired end-effector wrench based on the position error and virtual spring-damper dynamics.
2. **Joint Space Mapping & Gravity Compensation:** Translates the desired Cartesian wrench into joint torques (via the Jacobian transpose) and adds the model-based gravity compensation term to counteract the static loads.
3. **Discrete-Time Cascaded Control:** A digital PI control architecture that regulates the motor current (inner loop) and the SEA output torque (outer loop), incorporating sampling delays and saturation limits.

4. **Actuator Dynamics & SEA:** Models the motor, the inertia and efficiency of the gearbox, the viscous friction of the gearbox and the compliant Series Elastic Element (spring) that decouples the motor inertia from the link.
5. **3D Rigid Body Dynamics:** The physical robot arm, modeled using the 20-sim 3D Mechanics Editor.

This interconnection is visualised in the word bond graph below, illustrating the flow of information (desired states) and power (torques and velocities) through the system:

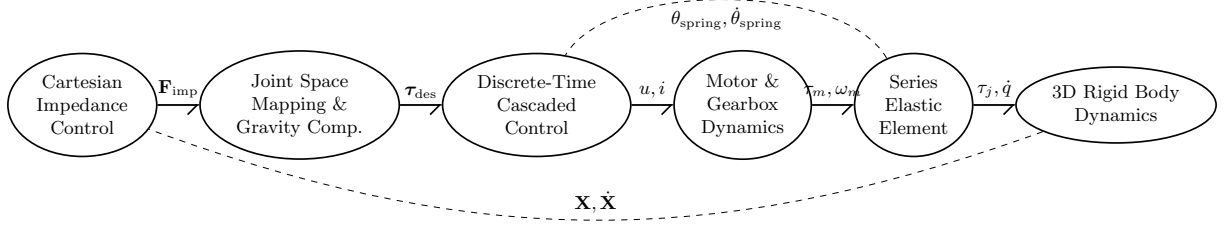


Figure 3: Word bond graph of the integrated 3-DoF robot arm system. Solid lines represent the primary forward flow of commands and power, while dashed lines represent the feedback signals required for closed-loop control.

3.1 Modelling choices

3.1.1 SEA Drivetrain and Friction Modeling

To capture the low-speed dynamics and steady-state error of the actuator, the gearbox total friction torque τ_{fric} reflected to the motor shaft is formulated as a combination of viscous and Coulomb friction:

$$\tau_{\text{fric}} = \alpha\omega + d_c \tanh(\gamma\omega) \quad (9)$$

where α is the viscous friction coefficient, d_c is the Coulomb breakaway torque, and $\gamma = 5 \text{ s/rad}$ provides a smooth numerical transition near zero velocity to maintain solver stability.

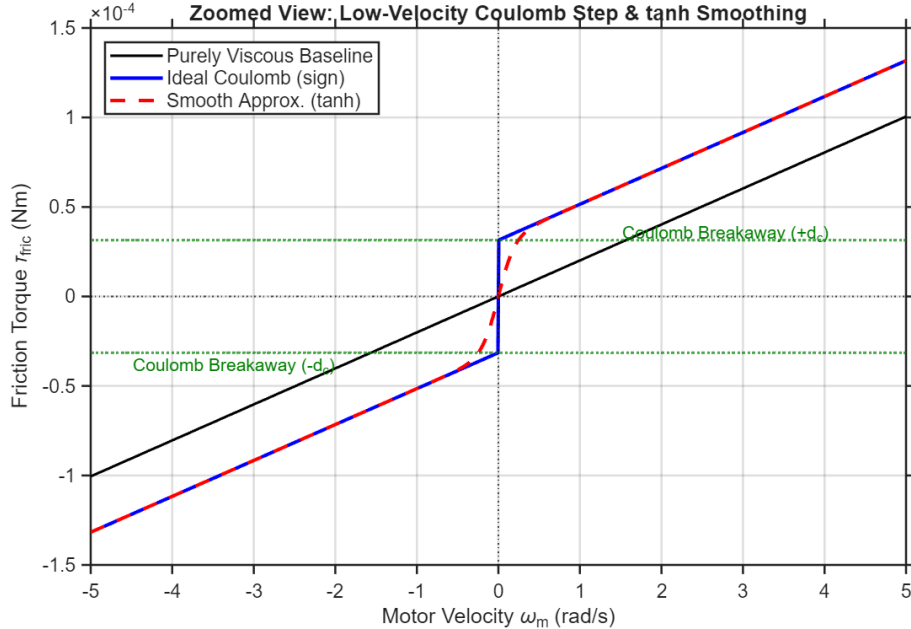


Figure 4: Coulomb and viscous friction approximation

The total internal torque lost to friction is constrained by the gearbox manufacturer specification of $\eta = 0.60$ efficiency [2]. At the nominal continuous operating speed ($\omega_{\text{rated}} = 694.3 \text{ rad/s}$), the total

friction torque accounts for the 40% efficiency deficit, yielding a total equivalent motor-side friction of $\tau_{\text{fric, total}} = 0.01396 \text{ Nm}$.

We approximate the output-side Coulomb friction as 5% of the nominal joint operating torque (0.2 Nm for the joint which requires least torque (second pitch joint)), resulting in 0.01 Nm. Reflected to the motor side through the gear ratio $i = 318$, the motor-side Coulomb parameter is:

$$d_c = \frac{0.01 \text{ Nm}}{318} \approx 0.0000314 \text{ Nm} \quad (10)$$

The remainder of the total friction loss is attributed to speed-dependent viscous effects. Isolating the viscous slope α results in:

$$\alpha = \frac{\tau_{\text{fric, total}} - d_c}{\omega_{\text{rated}}} = \frac{0.01396 \text{ Nm} - 0.0000314 \text{ Nm}}{694.3 \text{ rad/s}} = \boxed{2.006 \cdot 10^{-5} \text{ Nm} \cdot \text{s/rad}} \quad (11)$$

Because the estimated Coulomb friction is very small relative to the total 40% power loss, the vast majority of the friction at the rated speed is captured by the viscous term. This effectively lumps the load-dependent gear-mesh losses into the viscous coefficient. This simplification heavily favors viscous friction over Coulomb friction. During development, we observed that increasing d_c beyond this value prevented the motor from sufficiently overcoming static friction to initiate the necessary motion, even though theoretical torque requirements were satisfied. We were unable to identify the root cause within the available time. Because d_c is tuned for a 0.2 Nm joint load, it will underestimate the load-dependent friction for other joints on the robot that require up to 3 Nm. The total friction magnitude still accurately captures the $\sim 40\%$ power loss of the gearbox.

We use the hyperbolic tangent (tanh) function rather than the mathematically pure but discontinuous signum (sgn) function so the function remains differentiable everywhere. In the bond graph domain, this is implemented as a non-linear Resistive (R) element connected to the mechanical 1-junction of the motor shaft (FRICTION_GB in Figure 5).

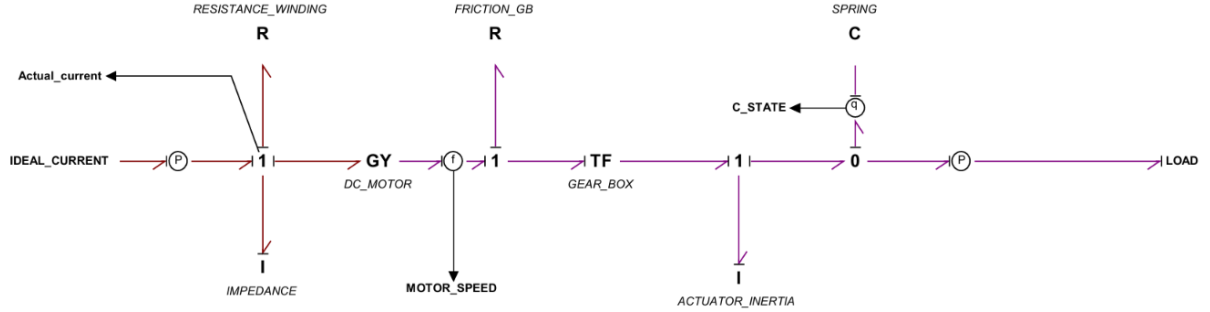


Figure 5: SEA Drivetrain Inside SEA_DRIVETRAIN sub-model, file: any of our .emx models with a SEA

3.1.2 SEA stiffness

The selected SEA stiffness was ($k_{SEA} = 100 \text{ Nm/rad}$). Taking the previously calculated worst-case gravity torque of (2.25 Nm), the corresponding elastic deflection of the SEA spring is

$$\theta_{SEA} = \frac{\tau}{k_{SEA}} = \frac{2.25}{100} = 0.0225 \text{ rad} \quad (12)$$

Assuming an effective lever arm from the pitch joint to the end-effector of approximately (0.5 m), this angular deflection corresponds to an approximate end-effector displacement of

$$\Delta x \approx L_{\text{eff}} \theta_{SEA} = 0.5 \cdot 0.0225 = 0.01125 \text{ m} \quad (13)$$

or roughly (1.1 cm). This was considered an acceptable value, since it provides enough compliance for the SEA while keeping the elastic deflection of the end-effector relatively small under the worst-case static gravity load.

3.1.3 Submodel Discretisation and Spring Dynamics

The model of the SEAs was discretised to interface with the digital controller. The physical plant is simulated using a fixed-step numerical solver with a step size small enough to capture the fast electrical and mechanical changes (10 kHz).

For the outer loop to remain stable and effectively regulate the robot's macroscopic impedance, the inner loop must be faster. By operating the inner SEA loops at 10 kHz, it is an order of magnitude faster than the 1 kHz outer arm controller. This ensures that the outer loop sees the actuator as a quasi-static, ideal torque source, preventing the fast dynamics of the inner loop from interfering with the whole-arm control.

The SEA introduces a two-inertia system (the reflected motor/gearbox rotor inertia J_m and the output link inertia J_l) connected by the spring stiffness k_{sea} . This configuration introduces a mechanical resonance frequency:

$$\omega_{res} = \sqrt{k_{sea} \left(\frac{1}{J_m} + \frac{1}{J_l} \right)} \quad (14)$$

For the SEA to behave as a controllable torque source, the inner control loop's operating bandwidth must be higher than ω_{res} to actively damp the spring and prevent mechanical oscillations. The achieved inner-loop bandwidth easily exceeds this physical resonance (which lies below 20 Hz depending on the joint configuration). This guarantees that the spring is tightly regulated by the inner loop.

Consequently, the whole-arm Cartesian impedance loop must have a bandwidth lower than the SEA's torque control bandwidth, but higher than the frequency of the reference trajectory.

This outer-loop bandwidth can be estimated by treating the Cartesian impedance law as a virtual mass-spring-damper system ($M_{eff}\ddot{x} + D\dot{x} + Kx = F_{ref}$). Using our chosen compliant parameters ($K = 50 \text{ N/m}$, $D = 3 \text{ N} \cdot \text{s/m}$) and estimating an effective end-effector mass of $M_{eff} \approx 2.3\text{--}3.5 \text{ kg}$ (based on the 2.22 kg physical arm mass and configuration-dependent inertia variations), the virtual natural frequency is:

$$f_n = \frac{1}{2\pi} \sqrt{\frac{K}{M_{eff}}} \approx 0.6\text{--}0.75 \text{ Hz} \quad (15)$$

This indicates that the outer impedance loop's effective bandwidth is roughly 0.6 Hz. This relatively low bandwidth directly dictates the maximum speed at which the robot can accurately track a trajectory. For a circular trajectory with radius R and frequency f , the end-effector speed is constant: $v = 2\pi f R$.

To avoid severe performance degradation, the trajectory frequency should remain safely below this natural frequency f_n .

For a circular path with $R = 0.1 \text{ m}$, this theoretical limit leads to a maximum end-effector speed of approximately $v_{limit} = 2\pi(0.5)(0.1) \approx 0.31 \text{ m/s}$. Attempting to track trajectories faster than this limit causes the system to act as a low-pass filter, resulting in the gradual amplitude attenuation and phase lag observed during our high-speed tracking tests.

3.1.4 Cascaded Control Architecture of the SEA

The SEA employs a cascaded control structure consisting of an outer impedance control loop and an inner torque/current control loop, implemented in discrete-time as shown in Figure 6.

Outer loop

The outer loop implements impedance control through proportional integral parameters plus a feed forward term. The desired torque τ_{des} is computed as:

$$\tau_{des}(t) = K_{ff} \cdot q_{ref}(t) + K_p \cdot (q_{ref}(t) - q(t)) + \tau_i \int_0^t (q_{ref}(\tau) - q(\tau)) d\tau$$

where K_{ff} is the feed forward gain, q_{ref} is the reference position, q is the measured position, τ_i is the integral gain and K_p the proportional gain.

Our current position comes out as a signal value from the SEA_DRIVETRAIN sub model where we sense which position is the SEA spring in and take it up in the model architecture, C_STATE in Figure 5. This

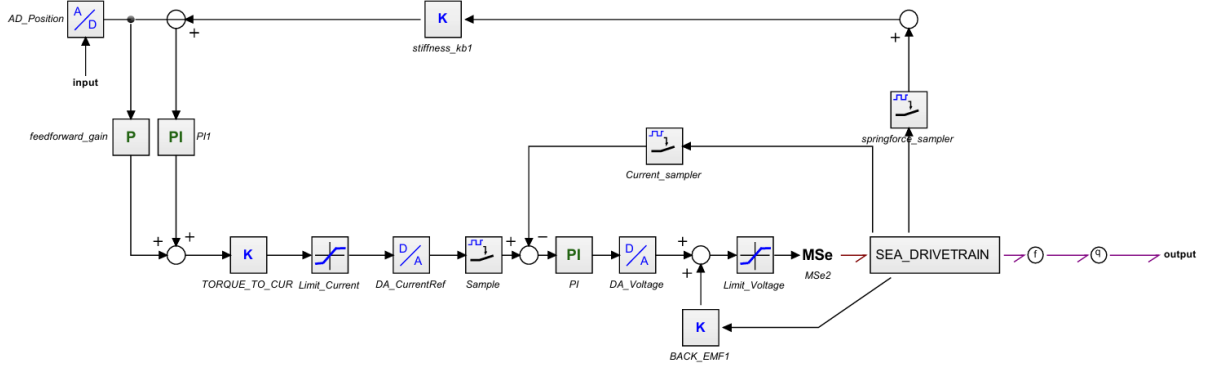


Figure 6: Cascaded control architecture for the SEA drivetrain with discrete-time sampling.

signal is multiplied by the spring stiffness (in the block *stiffness_kb1*: $1/cb1$, where $cb1$ is the spring compliance) to have the correct position.

The desired torque is converted to a current reference via the motor torque constant K_t and the gear ratio K_g :

$$i_{\text{ref}} = \frac{\tau_{\text{des}}}{K_t * K_g} \quad (16)$$

This equation converts the desired joint torque (τ_{des}) into the required motor current reference (i_{ref}). It accounts for the mechanical torque multiplication provided by the gear ratio (K_g) and the electro-mechanical conversion efficiency of the motor determined by its torque constant (K_t).

After this we apply a current limiter to ensure motor safety and realistic values, then we change to analog data via the D/A block and start sampling the signal to enter the inner loop.

Table 4: Outer Loop Control Parameter Values

	K_{ff}	K_p	τ_i	K_t	K_g
Value	1	2.8	0.18	0.0129	318

Inner loop

A discrete-time PI controller regulates the motor current, with anti-windup protection and saturation limits to respect the motor's continuous current rating. A Back EMF compensation is added just before a later voltage limiter that prevents over-tension. The controller output drives the motor as a controlled voltage source. The current to voltage conversion is not explicitly done like we have done in the outer loop with torque-to-current, instead it is carried out by the K_{p2} proportional gain following the $V = I * R$ law.

$$Voltage_{ref}[k] = (ang_{velocity} * K_{bEMF}) + K_{p2}(\alpha_{ref}[k] - \alpha[k]) + \frac{K_{p2} \cdot T_s}{\tau_I} \sum_{j=0}^k (\alpha_{ref}[j] - \alpha[j])$$

Table 5: Inner Loop Control Parameter Values

	K_{bEMF}	K_{p2}	τ_I
Value	0.0129	0.5	0.03

3.2 Impedance Control of the 3-DoF Robot Arm

In the 3-DoF robot arm Cartesian impedance control is used to regulate the movement of the end-effector. The control architecture, shown in Figure ??, consists of a Cartesian PD controller, force saturation (included in the controller block), Jacobian transpose mapping (in the force_to_torque block), and gravity compensation (also in the force_to_torque block).

The desired Cartesian force $\mathbf{F}_c \in \mathbb{R}^3$ is calculated using a proportional-derivative (PD) law:

$$\mathbf{F}_c = \mathbf{K} \mathbf{e}_p + \mathbf{D} \mathbf{e}_v \quad (17)$$

where $\mathbf{e}_p = \mathbf{p}_{\text{ref}} - \mathbf{p}$ and $\mathbf{e}_v = \dot{\mathbf{p}}_{\text{ref}} - \dot{\mathbf{p}}$ are the position and velocity errors, respectively. The diagonal stiffness and damping matrices are:

$$\mathbf{k} = \begin{bmatrix} 50 \\ 50 \\ 50 \end{bmatrix} \text{ N/m}, \quad \mathbf{d} = \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix} \text{ N} \cdot \text{s/m} \quad (18)$$

These stiffness values and damping values were decided based on manual tuning. But we made sure via validation and testing that if we applied a 2 N external force in the end effector the reaction of the robot was reasonable and realistic, therefore making the values reasonable to our model but also to reality.

These low values prioritise high compliance and soft interaction behavior. Due to this prioritisation the system is strongly underdamped. In a standard linear second-order system, the damping coefficient required to achieve a specific damping ratio ζ (e.g., critical damping where $\zeta = 1$) is dictated by the equation:

$$D_{\text{ideal}} = 2\zeta\sqrt{M_{\text{eff}}K} \quad (19)$$

where M_{eff} is the effective mass of the system. In the Cartesian impedance control of our system, however, the effective mass at the end-effector is not constant. It is defined by the configuration-dependent Cartesian inertia matrix $\Lambda(\mathbf{q}) = (\mathbf{J}(\mathbf{q})\mathbf{M}^{-1}(\mathbf{q})\mathbf{J}^T(\mathbf{q}))^{-1}$.

Because M_{eff} varies across the workspace, strictly following the ideal damping equation would require real-time gain scheduling of the damping matrix $\mathbf{D}$ based on the current joint configuration. To maintain passivity, numerical stability, and computational simplicity, we opted for the constant damping coefficient, which was found through manual tuning. For an estimated effective end-effector mass of $M_{\text{eff}} \approx 2.3\text{--}3.5$ kg, the damping ratio $\zeta = \frac{D}{2\sqrt{M_{\text{eff}}K}}$ falls between 0.11 and 0.14.

While this ensures the robot yields easily to external forces, the low amount of damping and low inertia of the final arm section mean the end-effector exhibits ringing when tracking sharp trajectory changes or reacting to sudden disturbances.

To ensure realistic interaction forces and protect the actuators, the Cartesian force magnitude is limited to $F_{\text{max}} = 50$ N. This saturation preserves the force direction while preventing excessive torques at the joints.

The limited Cartesian forces are mapped to joint torques using the Jacobian transpose (derived in Section 2.3):

$$\boldsymbol{\tau}_{\text{imp}} = \mathbf{J}(\mathbf{q})^T \mathbf{F}_{\text{limited}} \quad (20)$$

Gravity compensation torques $\boldsymbol{\tau}_g(\mathbf{q})$ are added to counteract the arm's weight:

$$\boldsymbol{\tau}_{\text{des}} = \boldsymbol{\tau}_{\text{imp}} + \boldsymbol{\tau}_g(\mathbf{q}) \quad (21)$$

The total desired torque $\boldsymbol{\tau}_{\text{des}}$ is then sent to the actuator control layer.

Joint positions and velocities are sampled using zero-order hold (ZOH) elements, and the computed torques are held constant between samples. Since the impedance law is purely algebraic (no internal dynamics), no discretisation of the control equations themselves is required—only the I/O signals are sampled.

4 Tuning strategy

For the tuning of the robot arm control, we followed a progressive and systematic strategy. The objective was to avoid tuning the complete system at once, since the interaction between the Cartesian impedance controller, the SEA compliance, the cascaded PI controllers, and the full 3D rigid-body dynamics can easily lead to coupled instabilities.

First, each SEA was tuned individually using the simulations presented in Assignment 7 as a reference. In particular, we used the gravity-compensated disturbance test and the circular trajectory with a velocity of 8 rad/s and radius $R=0.1$ m. These simulations were used to generate sinusoidal approximations of the expected torque dynamics, with similar amplitude and frequency, in order to calibrate each SEA separately before integrating it into the complete robot arm. For these isolated tests, the output of the SEA was connected to an ideal zero-flow source through a 1-junction. This allowed the load side to remain fixed while the torque tracking behaviour of the SEA could be evaluated independently.

The tuning was carried out from the inner loop to the outer loop. First, we ensured that the motor current could accurately track its reference using the inner current loop. After achieving acceptable current tracking, the outer torque loop was tuned to make the SEA output torque follow the desired torque reference. At both levels, the goal was not only to reduce the tracking error, but also to avoid instabilities, excessive ripple, and saturation of the current or voltage limits.

Once each SEA behaved correctly in isolation, the ideal effort sources in the complete robot model were progressively replaced by the SEA models. The integration was done joint by joint: first the distal pitch joint (X2), then the proximal pitch joint (X1), and finally the yaw joint (Z). This progressive replacement made it possible to identify which joint was responsible for any instability or degradation in tracking, instead of introducing all actuator dynamics at once.

During this integration process, we observed that the parameters obtained from isolated SEA tuning could not be used directly in the full robot model. When the SEAs were connected to the robot arm, the coupled link dynamics, gravity loading, actuator inertia, gearbox dynamics, and SEA compliance changed the behavior of the system significantly. Therefore, the control parameters had to be adjusted again during the integration stage. In practice, the final tuning was the result of a compromise between trajectory tracking accuracy, torque tracking quality, current ripple, and stability of the coupled robot-SEA system.

The original circular trajectory from Assignment 7 was also found to be too aggressive once realistic SEA dynamics were included. Although the task with a velocity of 8 rad/s and radius 0.1 m was useful for stressing the isolated SEA models, it generated excessive torque demands and oscillatory behavior in the full integrated robot. Therefore, the final circular trajectory was made more conservative by reducing the velocity to 1 rad/s, while keeping the same radius.

In addition, the simulation was modified so that the robot maintained the initial position for 5 seconds before starting the motion. This settling phase ensured that the current, torque, and spring states reached a consistent value before the trajectory began. Since the robot is already loaded by gravity at the initial configuration, the SEA springs required an initial pre-load. For the initial end-effector position $x=0$ m, $y=0.35$ m, $z=0.35$ m, the spring initial deflections were set to 17.5 mrad for joint (X1) and 2.7 mrad for joint (X2). No equivalent gravity pre-load was required for the yaw joint (Z), since gravity does not generate a static torque around the vertical yaw axis.

After these changes, the integrated model was evaluated using current tracking, SEA torque tracking, and Cartesian trajectory tracking. The final controller settings were selected based on the complete integrated behavior rather than on the isolated SEA tests alone. The selected tuning provided acceptable tracking while keeping the ripple and actuator demands within reasonable limits.

5 Results

Two tests were performed to validate the model: holding a fixed position under external disturbances and following a circular trajectory.

5.1 Disturbance Test:

In the disturbance test, a smooth pulse force with a magnitude of 2 N was applied from $t=6$ s to $t=8$ s. The test was repeated by applying the disturbance independently in the X, Y, and Z directions.

With a Cartesian impedance stiffness of 50 N/m, the expected quasi-static displacement in the direction of the applied force is

$$\Delta x = \Delta y = \Delta z = \frac{F}{k} = \frac{2}{50} = 0.04 \text{ m} = 4 \text{ cm} \quad (22)$$

Figures 7, 8 and 9 show that the displacement was of this order of magnitude. However, the displacement did not remain perfectly aligned with the direction of the applied force. In particular, Figures 8 and 9 show that a disturbance in Y also generated motion in Z, and vice versa. This coupling is expected from the kinematic structure of the robot. Both Y and Z positions are mainly determined by the two pitch joints. Therefore, any disturbance that excites these pitch joints affects both coordinates simultaneously.

The disturbance in X, shown in Figure 7, produced a smaller secondary displacement in Y. This is because motion in the X direction is mainly associated with the yaw joint. A yaw rotation moves the end-effector along an arc in the horizontal plane, so a displacement in X can also introduce a smaller change in Y. In contrast, the coupling with Z is weaker because the yaw joint rotates around the vertical axis and does not directly change the height of the end-effector. These results show that, although the Cartesian stiffness matrix is diagonal, the closed-loop response of the complete robot is not perfectly decoupled due to the nonlinear kinematics, SEA compliance, actuator dynamics, and gravity compensation.

On the other hand, this test also showed that, after approximately 3 seconds of current and torque controller settling, the robot arm stabilized around the reference position. This indicates that the gravity compensation was correctly implemented, since the arm was able to maintain the static pose without a significant steady-state error once the initial transients had disappeared.

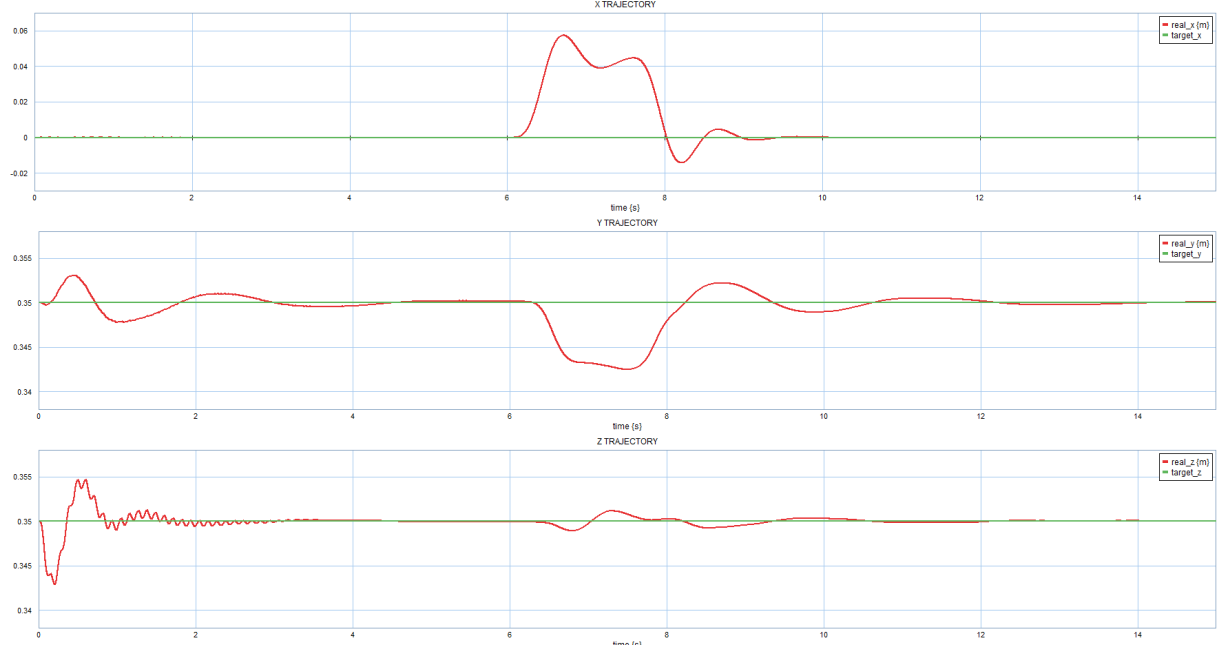


Figure 7: Trajectories reference vs real trajectories in a static position with a 2N smooth pulse disturbance in X from second 6 to 8. Notice the small scale in the Z trajectory.

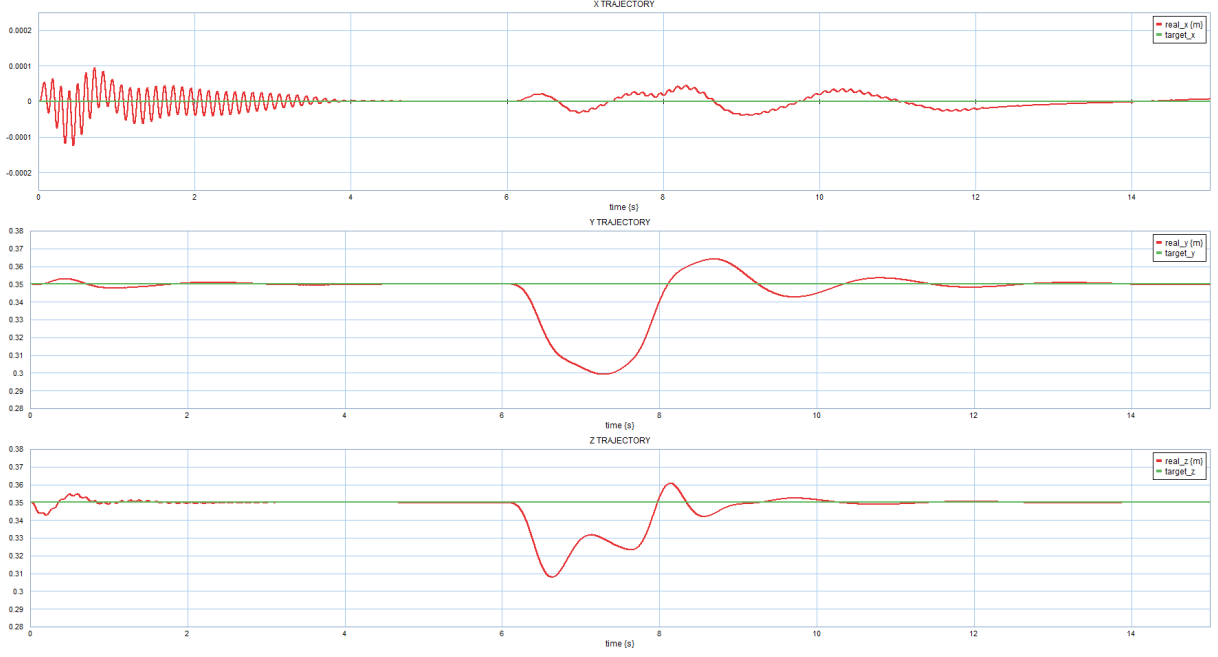


Figure 8: Trajectories reference vs real trajectories in a static position with a 2N smooth pulse disturbance in Y from second 6 to 8. Notice the small scale in the X trajectory.

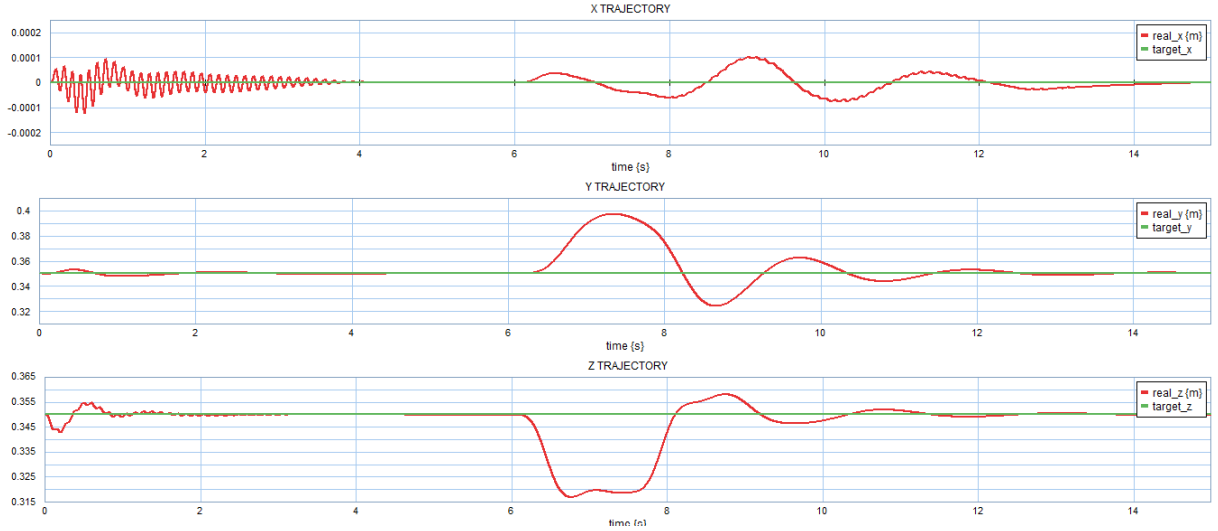


Figure 9: Trajectories reference vs real trajectories in a static position with a 2N smooth pulse disturbance in Z from second 6 to 8. Notice the small scale in the X trajectory.

5.2 Circle trajectory

For the circular trajectory, the end-effector followed the reference path closely in all three Cartesian coordinates, as shown in Figure 10. However, it is important to note that the final trajectory was intentionally conservative. The angular velocity was reduced compared with the original task in order to avoid exciting the coupled SEA and robot-arm dynamics and to preserve stability in the fully integrated model.

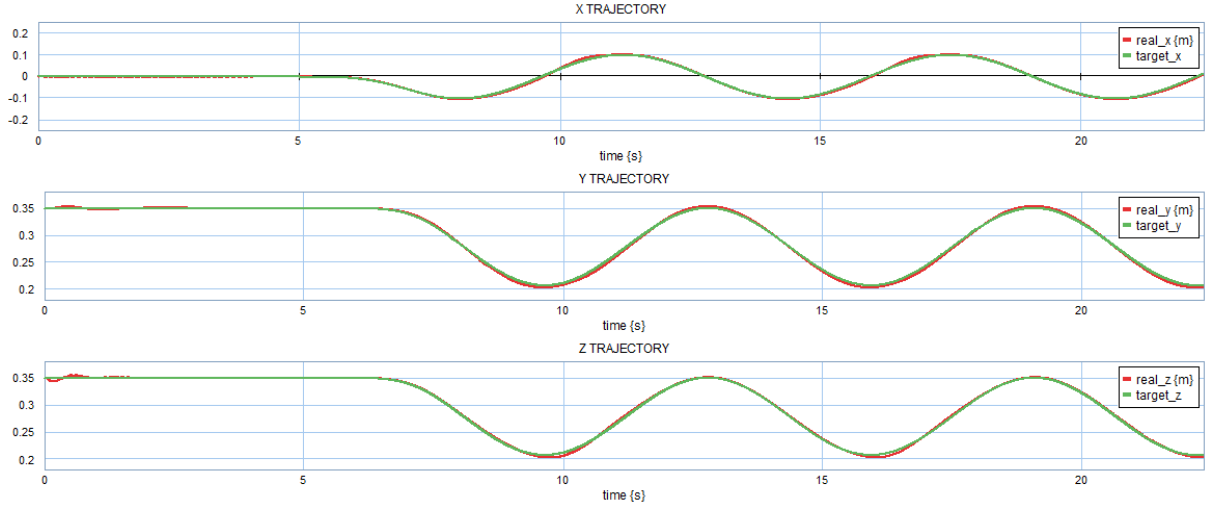


Figure 10: Trajectories reference vs real trajectories for each actuator in the final model.

The current tracking for each actuator is shown in Figure 11. Overall, the actual current follows the current reference well. At the beginning of the simulation, small oscillations are visible while the current loop settles from the initial conditions. These oscillations remain limited in amplitude and decay during the initial holding phase. However, after approximately 15 s, the current signal of the distal pitch joint (X2) starts to show higher-frequency oscillations. This behaviour is undesirable, since it indicates that the actuator is being excited by faster dynamics during the circular motion.

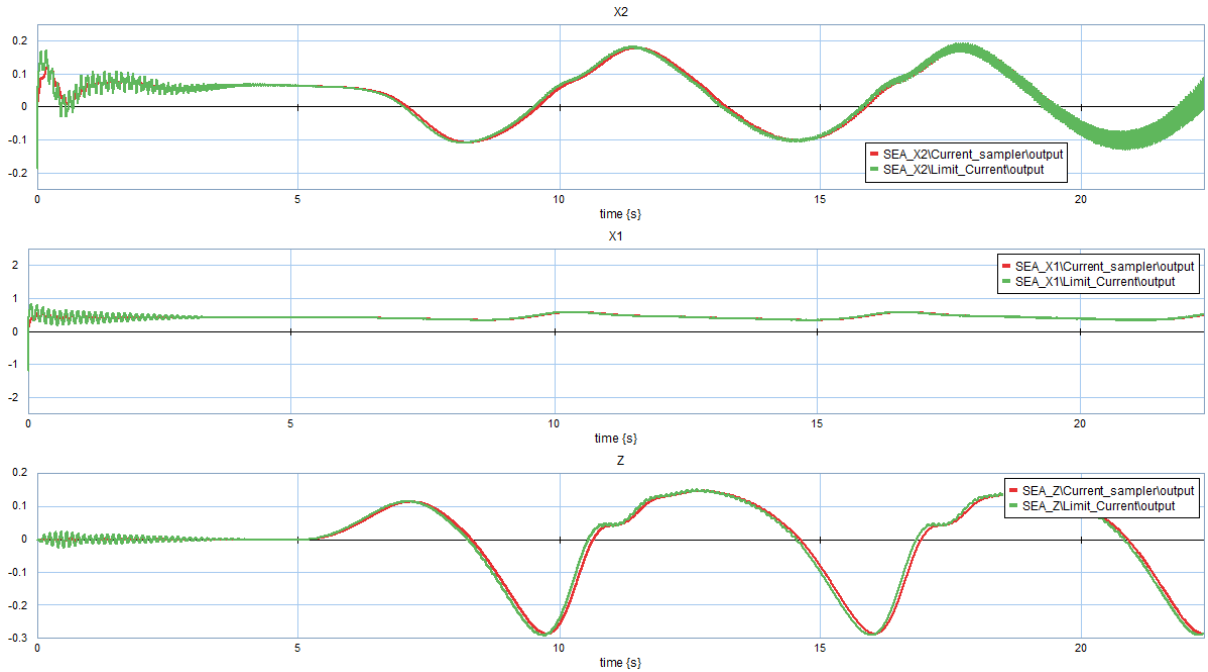


Figure 11: Current reference vs supply for each actuator in the final model.

A similar behaviour can be observed in the torque tracking results shown in Figure 12. The applied torque follows the torque reference reasonably well for all joints, which confirms that the cascaded current and torque control structure is functioning as intended. The proximal pitch joint (X1) presents the largest torque amplitude, which is expected because this joint carries the inertia and gravitational load of the

distal links and actuators. After approximately 15 s, the torque response of (X2) also presents an increase in oscillation frequency, consistent with the current response observed in Figure ??.

The correlation between current and torque confirms that the electromechanical conversion and drivetrain model are behaving consistently: changes in motor current are reflected in the generated actuator torque. Nevertheless, the high-frequency oscillations observed in (X2) indicate a limitation of the final tuning, most likely related to the interaction between the outer torque loop, the SEA compliance, and the low inertial load of the distal joint.

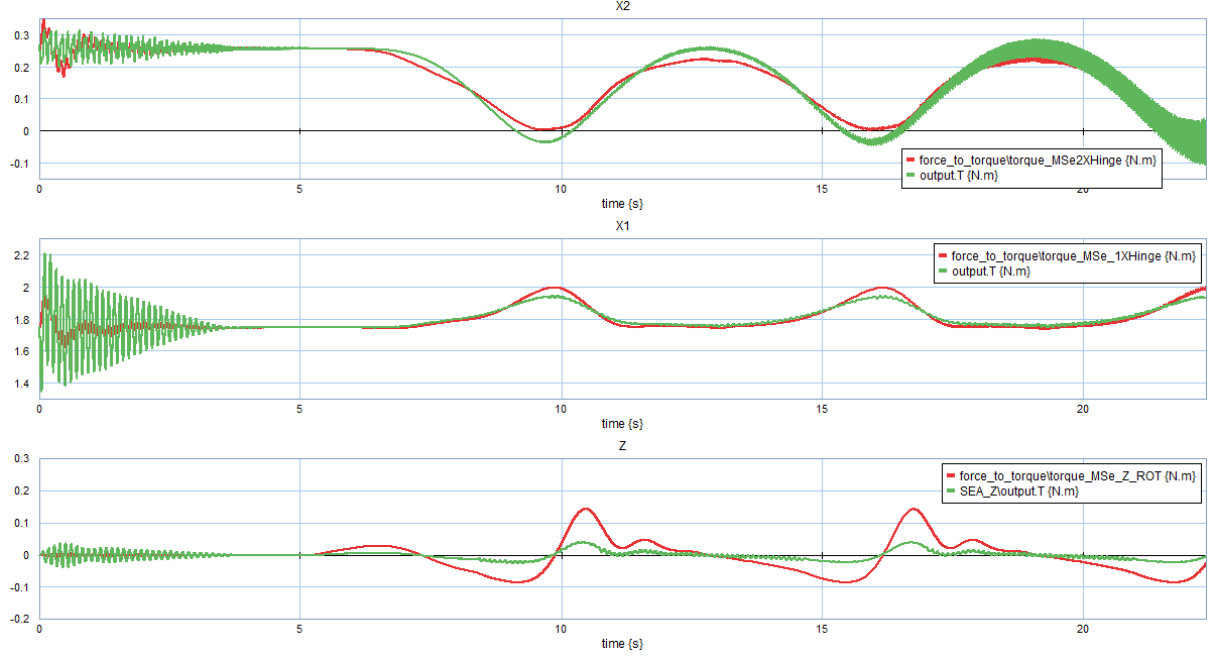


Figure 12: Torque reference vs supply for each actuator in the final model.

To further analyse the internal behaviour of the SEA controllers, the contributions of the feedforward and PI terms in the outer torque loop were plotted in Figure 13. These plots show how the desired torque command is generated from both the direct feedforward contribution and the PI correction. The feedforward term provides the main torque contribution during smooth trajectory tracking, while the PI term compensates for the remaining torque error. The oscillatory behaviour observed in the actuator torques is also visible in these control signals, confirming that the outer torque loop directly influences the current and torque responses of the SEAs.

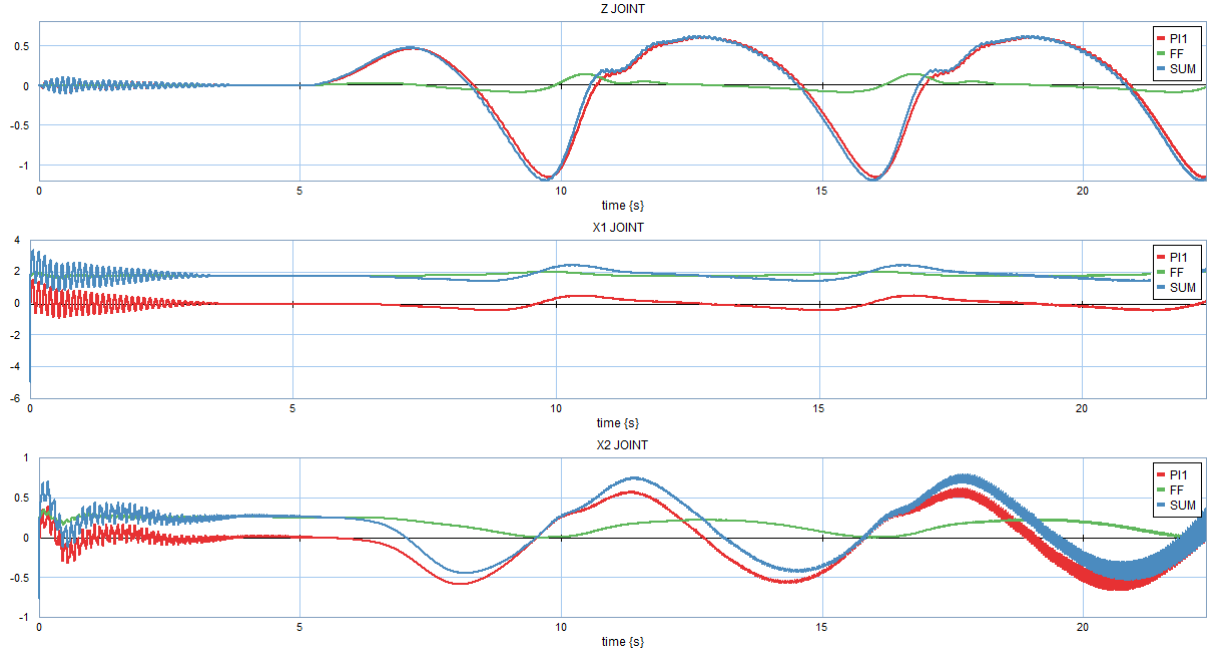


Figure 13: Summation blocks inputs and outputs FF and PI1 in the final model.

The inner current loop contributions are shown in Figure 14. The Back-EMF compensation provides the dominant contribution when motor speed increases, while the PI controller corrects the residual current tracking error. The start-up oscillations are again visible, especially in joint (X1), but no high-frequency oscillations are observed after 15 s. This suggests that the later oscillations are not mainly caused by the current controller itself, but are more likely introduced by the outer torque loop or by the coupled SEA-robot dynamics.

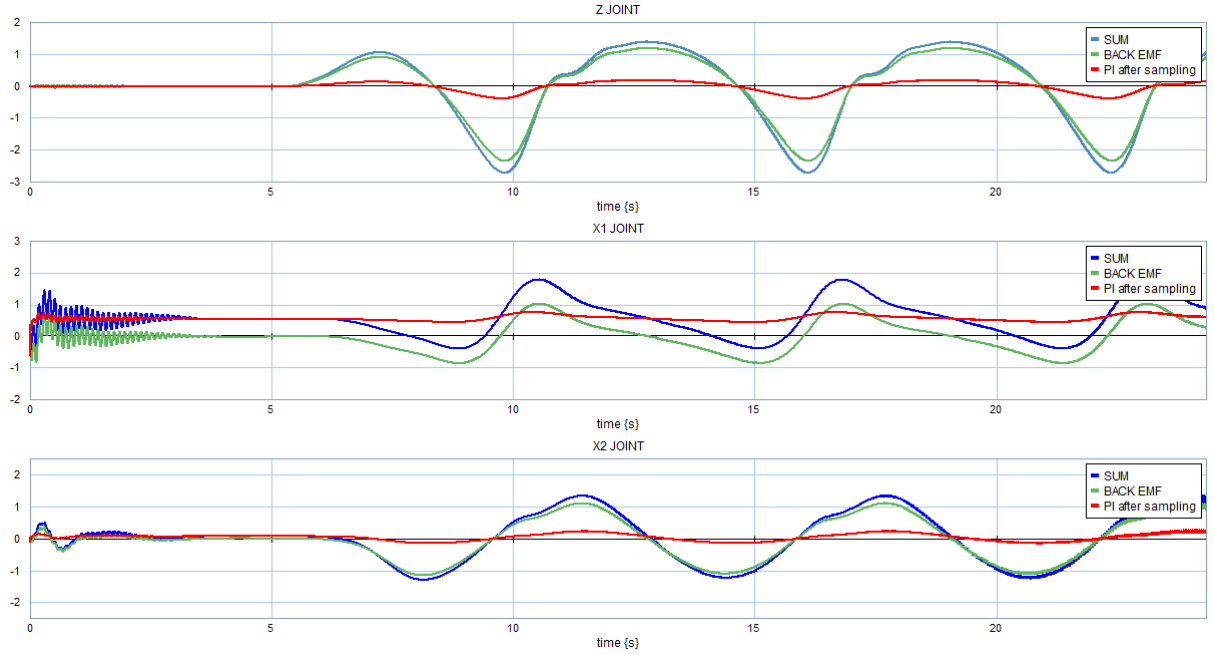


Figure 14: Summation blocks inputs and outputs for Inner Loop PI and Back EMF in the final model.

6 Discussion & Conclusion

Overall, the model can follow a conservative circle trajectory and reacts with an expected behavior to small disturbances. Nevertheless, there are things that did not go as planned.

During actuator selection in Section 2.2, the gear ratio N was increased significantly, in order to extract sufficient output torque from the new motor. While Table 3 confirms this choice satisfies the torque requirement, we found while testing that the resulting system contained one unstable joint. The likely cause was how the increased N amplifies the effect of the Coulomb friction within the torque loop.

The Coulomb friction in our model is reflected on the motor side, but its effect on the output-side is scaled by the gear ratio once it passes through the transmission. Raising N and directly increasing the magnitude of the friction disturbance felt in the torque loop, without any change to D_c itself. At the same time, the reflected motor inertia scales with N^2 , which through the resonance relation pulls the SEA's resonance frequency down, reducing the frequency range in which the inner loop still has phase margin.

These effects add to each other as coulomb friction is steepest near zero velocity, exactly where the inner loop operates during the start of a motion. As N increases, this nonlinearity is amplified and felt within an already reduced bandwidth, so it behaves less like a small disturbance and more like a large source of phase lag near the loop's crossover frequency. This causes a loss of phase margin at low velocity that scales with N . friction that was negligible at a lower gear ratio becomes large enough after amplification, to destabilise the loop.

This also explains the joint dependence of the problem, consistent with our integration order in Section 5: instability appeared specifically on the joint with the smallest inertial load. Since the sensitivity at the spring scales inversely with inertia, the same amplified friction has proportionally less effect on joints with larger reflected load inertias, even under the same increased N . This suggests the instability resulted from the gear ratio amplifying an existing friction nonlinearity enough to destabilise the torque loop, particularly on the joint least able to absorb the disturbance.

This is a good example of a well verified model losing stability as realistic non-ideal elements are introduced to it. the piecemeal development and integration approach allowed for the isolation of the sources of these instabilities. despite not all sources of instability having been fully addressed the system achieves a stable trajectory and quick recovery from disturbances within a conservative limits implemented during tuning. additionally the model also exposed hardware limits most notably the torque margin for heavy lifting and the friction-induced instability at high gear ratios, that would need to be addressed before the system could be considered ready for a realisation.

7 AI Statement

In the writing of this report the following Large Language Models were used to aid in linguistic quality control as well as formatting for LaTeX:

- Qwen AI
- ChatGPT, OpenAI

All work was thoroughly checked and assessed for complete alignment with the vision of the authors.

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